

Homework 6 (RNNs including LSTM and BRNN)

Section 1: Multiple-Choice Questions (MCQs)

1. Why do standard Multi-Layer Perceptrons (MLPs) struggle with tasks like binary addition on sequences of varying lengths?
 - A) They require a predefined, fixed number of input units.
 - B) The weights applied at the beginning of the sequence are different from those at the end, preventing the model from generalizing a rule (like carrying a '1').
 - C) They cannot use activation functions on binary numbers.
 - D) Both A and B.

2. During the training of an RNN Language Model, what is "Teacher Forcing"?
 - A) Punishing the model with a higher learning rate when it makes a mistake.
 - B) Feeding the model's own predicted word back into it as the next input.
 - C) Feeding the *actual* correct next word from the training corpus as the input for the next time step, regardless of what the model predicted.
 - D) Forcing the hidden state to reset to zero after every sentence.

3. In the PyTorch data construction example for Autoregressive Models, what does the `tau` parameter represent?
 - A) The learning rate of the model.
 - B) The window size, meaning the number of previous time steps used as features to predict the next time step.
 - C) The total number of epochs the model will train for.
 - D) The threshold used for gradient clipping.

4. In a Long Short-Term Memory (LSTM) network, which component is primarily responsible for deciding what new information should be stored in the cell state?
 - A) The Forget Gate
 - B) The Input Gate combined with the Candidate Memory
 - C) The Output Gate
 - D) The Softmax layer

5. How does a Bidirectional Recurrent Neural Network (BRNN) differ fundamentally from a standard, unidirectional RNN?

A) It processes the sequence in both forward and backward directions to utilize past and future context simultaneously.

B) It restricts the network to only predict the previous sequence step instead of the next one.

C) It limits the gradient backpropagation to only move backward through the hidden layers, not through time.

D) It requires the input data to be duplicated and fed into the network twice at the same time step.

Section 2: True/False Questions

6. True or False: In an unrolled Recurrent Neural Network, the weight matrices (e.g., the hidden-to-hidden weights) are updated and change their values at every single time step within a sequence.

7. True or False: The "vanishing gradient" problem makes it difficult for an RNN to learn long-term dependencies because the gradients become too small to meaningfully update the weights for earlier time steps.

8. True or False: To solve the exploding gradient problem (which can cause NaN errors in your network), we use a technique called Gradient Clipping, which shrinks the gradient vector if its norm exceeds a specific threshold.

Section 3: Short Answer & Code Understanding

9. The Netflix Example: Briefly explain the "Netflix rating system" example (Koren, 2009) shown in the lecture. Why does this highlight the importance of treating data as a sequence rather than a static dataset?

10. LSTM Mathematics: Look at the core equation for updating the cell state (memory) in an LSTM: $C_t = F_t \odot C_{t-1} + I_t \odot \tilde{C}_t$. Briefly explain what the two halves of this addition are doing in plain English.

11. PyTorch Sequence Slicing: Look at the following Python logic based on the slides on RNNs for creating the data iterator:

```
features = [self.x[i: self.T-self.tau+i] for i in range(self.tau)]
```

```
self.labels = self.x[self.tau:].reshape((-1, 1)).
```

If our raw sequence data x is [10, 20, 30, 40, 50, 60] and τ is 3, what will be the features array and label for the very first training example?

Answers

Section 1: MCQs

1. **D** - MLPs require fixed input sizes and do not share weights across different positions in the sequence.
2. **C** - Teacher forcing means using the ground-truth previous token as the input for the next step during training.
3. **B** - τ defines the sliding window size of past observations used to predict the future.
4. **B** - The Input Gate decides if the new Candidate Memory should be added to the cell state.
5. **A** - BRNN is designed exactly for this purpose.

Section 2: True/False

6. **False** - The defining feature of an RNN is *parameter sharing*. The exact same weight matrices are applied at every time step.
7. **True** - Vanishing gradients prevent the network from linking a word at the end of a long paragraph to a context clue at the beginning.
8. **True** - Gradient clipping scales down the gradient step to prevent massive, destructive parameter updates.

Section 3: Short Answer & Code

9. **Answer:** If you look at Netflix movie ratings purely by the age of the movie, they seem to go up over time due to selection bias. However, if plotted by *date*, there is a sudden, massive jump in scores around day 1500 because Netflix fundamentally changed its rating UI/system. If a model doesn't understand the chronological sequence of events, it cannot account for this sudden behavioral shift.
10. **Answer:**

$F_t \odot C_{t-1}$: The Forget Gate F_t is multiplied by the old memory C_{t-1} to scale down or "forget" information we no longer need.

$I_t \odot \tilde{C}_t$: The Input Gate I_t scales the new Candidate Memory $\tilde{C}_t$ to decide how much of the new information should be "added" into the long-term memory state.

11. **Answer:** Features: [10, 20, 30] Label: [40]

(Explanation: With a window size of 3, the model looks at the first three numbers to predict the fourth).